



The Step You Can't Picture

Self-Reported Step Vividness Predicts When Assemblers Trade the Diagram Booklet for Video

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Motivation

Flat-pack manuals go wordless by design, on the assumption a diagram carries a step as reliably as a sentence would. Little is known about when — or why — an assembler gives up on the panel sequence and opens a video the manual never mentions.

Research questions

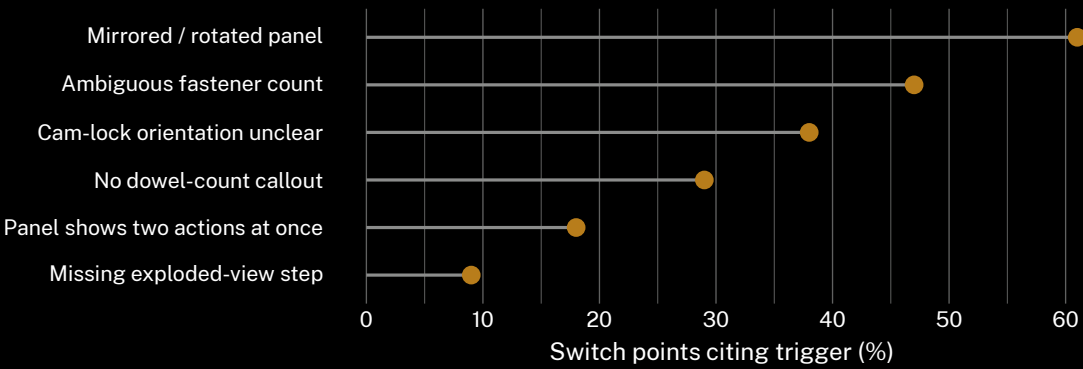
- Whether self-reported step-vividness predicts how many panels an assembler completes before switching to video.
- Which diagram features — mirrored orientation, ambiguous fastener counts, unlabelled cam-lock direction — precede a switch.
- Whether switching, once made, buys fewer errors or just a different kind of confidence.

Study design

- n = 214 assemblers intercepted mid-build at share-house move-ins and flat-pack resale stalls; vividness of the next step self-rated (1–5) before each panel; 20–45 min sessions, 5-month field window.
- Mixed-effects logistic regression: panel-level switch-to-video ~ vividness rating + panel step-count + fastener-type match, participant as random intercept.
- Kit model and manual edition held constant within each observed build; vividness ratings logged panel-by-panel, blind to the switch coding.

Findings

Switch-to-video odds rise sharply once reported vividness falls below the scale midpoint. Mirrored-panel exposure predicts a switch regardless of how vividly the step was pictured.



Mirrored-panel confusion was cited at 61% of logged switch points over the field window (n = 214), the most-cited of six coded triggers and 14 points clear of ambiguous fastener counts.

Implications & where next

Vividness looks like a gate on persistence, not a proxy for kit difficulty — low-vividness assemblers switch even on panels coded objectively simple. Mirrored panels concentrate the switching far more than step count alone, hinting a redrawn panel could out-perform a bundled video. Switchers report fewer errors, not faster completion. Next: a pre-registered trial randomising panel mirroring to separate the diagram effect from the vividness effect.

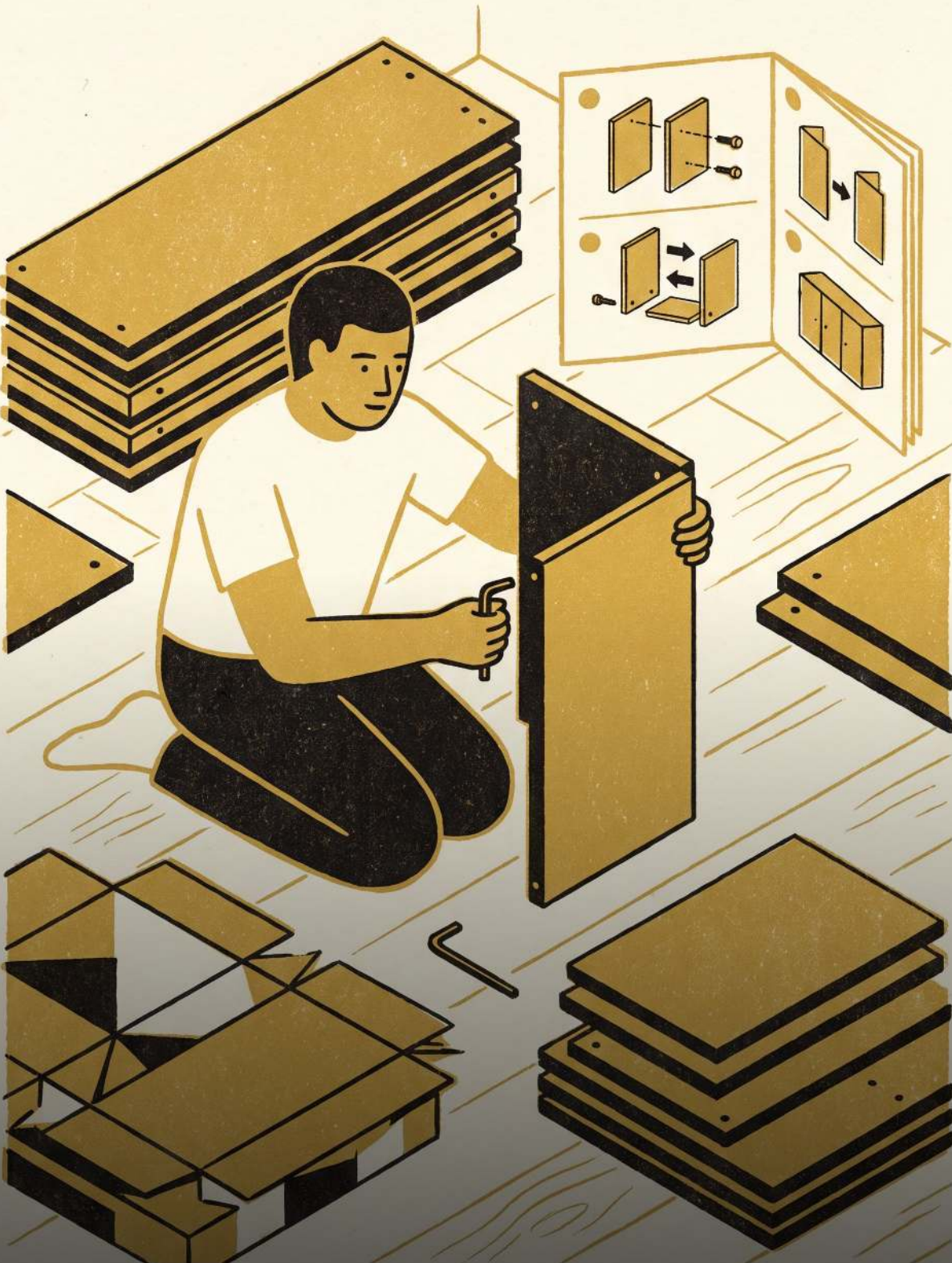
Selected references

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Ethics: informal-observation waiver 2026/031; diagram booklets and offcuts returned to the host kit after logging.

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“By panel six, most eyes have left the booklet.”

61% of logged switch points cite a mirrored or rotated panel (n = 214, 2026).